

Trainings ▾

Preparatory Steps

Remove the front drive belt. Refer to Procedure 008-002 (</qs3/pubsys2/xml/en/procedures/100/100-008-002.html>).

Remove the vibration damper/crankshaft pulley. Refer to Procedure 001-052 (</qs3/pubsys2/xml/en/procedures/100/100-001-052.html>).

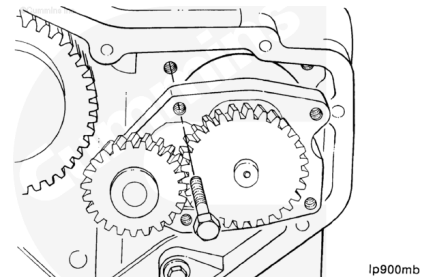
Remove the front gear cover. Refer to Procedure 001-031 (</qs3/pubsys2/xml/en/procedures/100/100-001-031.html>).



Remove

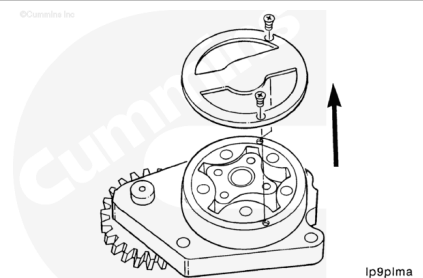
Remove the four oil pump mounting capscrews.

Remove the oil pump from its bore in the cylinder block.

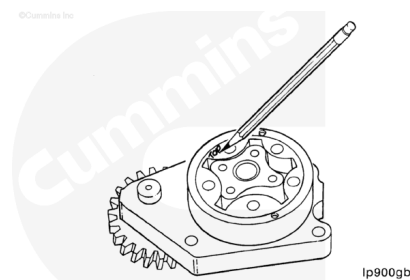


Disassemble

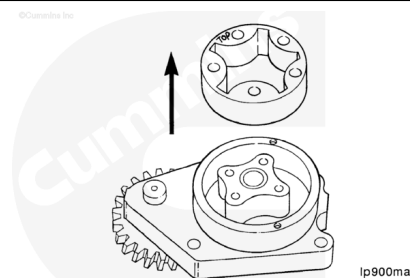
Remove the oil pump back plate.



Mark "TOP" on the gerotor planetary.



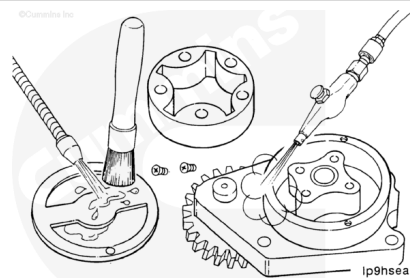
Remove the gerotor planetary.



Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Some solvents are flammable and toxic. Read the manufacturer's instructions before using.

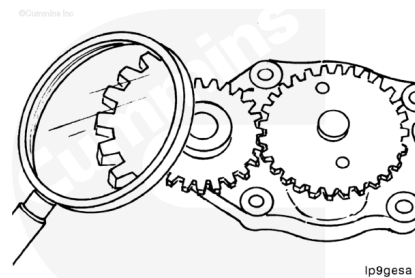
⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause bodily injury.

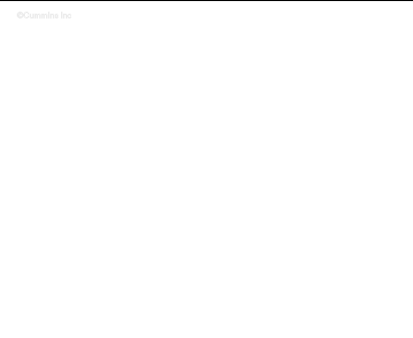
Clean all of the oil pump parts in solvent.

Dry all of the oil pump parts with compressed air.

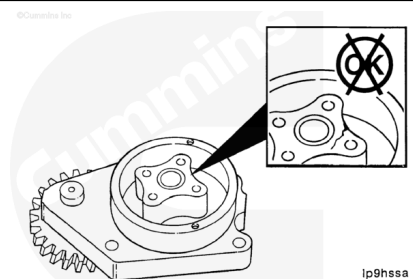
Inspect the lubricating oil pump gears for chips, cracks, or excessive wear.



Inspect the gerotor planetary for excessive wear or damage.

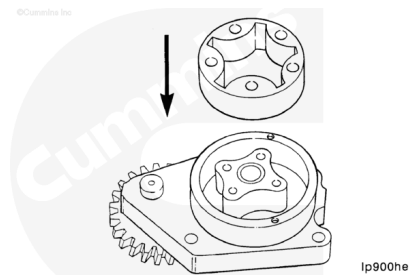


Inspect the lubricating oil pump housing and gerotor drive for damage and excessive wear.



Assemble

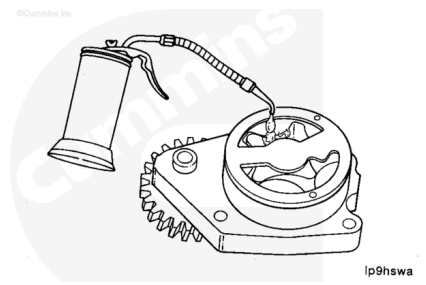
Install the gerotor planetary with the top mark in the correct position.



Install

⚠ CAUTION ⚠

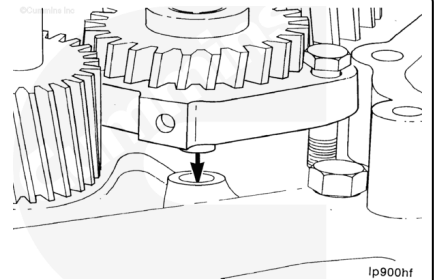
Failure to fill the pump with oil during installation can result in a slow prime at initial engine start-up, resulting in severe engine damage.



Lubricate the lubricating oil pump with clean engine oil.

⚠ CAUTION ⚠

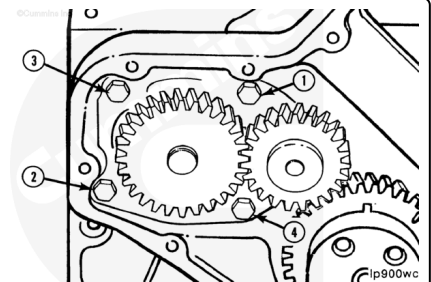
Make sure the idler gear pin is installed in the locating bore in the cylinder block. This can damage the pump.



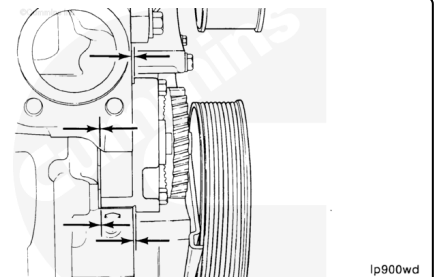
Install the lubricating oil pump.

Tighten the oil pump mounting capscrews in the sequence shown.

Torque Value: 24 n•m [18 ft-lb]

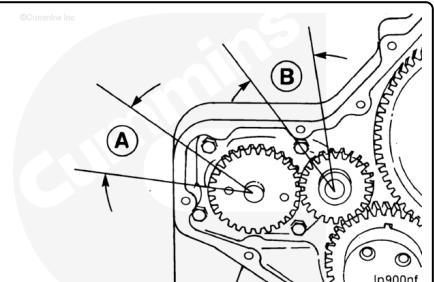


Note : The back plate on the oil pump seats against the bottom of the bore in the cylinder block. When the lubricating oil pump is correctly installed, the flange on the oil pump will not touch the cylinder block.



Measure the oil pump gear backlash.

	mm		in
A	0.076	MIN	0.003
	0.330	MAX	0.013
B	0.076	MIN	0.003
	0.330	MAX	0.013



Note : If the adjoining gear moves when you measure the backlash, the reading will be incorrect.

Finishing Steps

Install the front gear cover and gasket. Refer to Procedure 001-031 (</qs3/pubsys2/xml/en/procedures/100/100-001-031.html>).

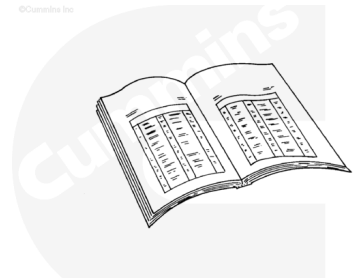
Install the front cover access cap and seal.

Install the crankshaft pulley/vibration damper. Refer to Procedure 001-052 (</qs3/pubsys2/xml/en/procedures/100/100-001-052.html>).

Install the drive belt. Refer to Procedure 008-002 (</qs3/pubsys2/xml/en/procedures/100/100-008-002.html>).

Fill the oil pan. Refer to Procedure 007-037 (</qs3/pubsys2/xml/en/procedures/100/100-007-037.html>).

©Cummins Inc.



ck800wa